Title:

Case study on Amazon EC2 to learn about Amazon EC2. Amazon EC2 Elastic Compute cloud is a central part of Amazon.com's cloud computing platform, Amazon Web Services. How EC2 allows users torrent virtual computers on which to learn run their own computer applications.

Case Study: Amazon EC2.

Abstract:

- Amazon Elastic compute cloud (EC2) forms a central part of Amazon.com's cloud computing platform, Amazon Web Services (AWS), by following users to rent virtual computers on which to run their own computer applications.
- EC2 encourages scalable development of applications by providing a web service through which a user can robot boot an Amazon Machine Image (AMI) to configure a virtual machine, which Amazon calls an "Instance", containing any software desired.
- A user can create, launch and terminate server-instances as needed, paying by the second for active servers - hence the term "elastic" EC2 provides users with control over the geographical location of instances that allows for latency optimization and high levels of redundancy.
- In November 2010, Amazon switches as its own retail website to use EC2 and AWS.

KEYWORDS:

- Introduction

- Architecture

Introduction:

EC2 uses Simple Storage Services (S3) for Storage of data. Users can hire suitable amount CPU power, Storage and memory without any upfront commitment. Users can control the entire Software Stack from kernel upwards.

Architecture:

The architecture has two components.

- S3
- EC2

S3 and EC2 is used for Storage and computing purposes.

Simple Storage Service:

- S3 can be through as a globally available distributed hash value table with high-level access control. Data is stored in pairs of name and value.
- Names are like UNIX file names and the value can be object having size up-to 5GB with up-to 4K of metadata for each object. Into the global namespace must fit all objects in Amazon's S3. This namespace contain a "bucketname" and an "object name".
- An AWS (Amazon Web services) account can have maximum of 100 buckets. Bucket names are like user names in traditional email account and provided by Amazon on first come first serve basis. This namespace Data to S3

can be sent by SOAP based API or with raw HTTP "PUT" commands.

- Data can be recovered using SOAP HTTP or BitTorrent. While using BitTorrent the S3 system operates as both tracker and the initial seeder. There are also some tools available which enables the users to view S3 as a remove file system.
- Upload download rate from and to S3 is not that much exiting one developer from Germany reported experiencing 10-100kBps. This rate can go upto 1-2MBps on the higher side depending on the time of the day.
- Although the speed is not that much fascinating it is good enough for delivering web objects and for backup purposes although for doing computation it is not suitable. Amazon S3 has a very impressive support for privacy, integrity and short term availability.
- Long term availability is unknown as this depends on the internal commitment of Amazon data centers. Data privacy can be obtained by encrypting the data to be stored. But this encryption is to be done by the user before storing the data in S3.
- One can use SSL with HTTPS to connect to S3 for more security but this uses of SSL increases upload/download time also. Data integrity can be achieved by checking end to end MD5 checking. When an object is stored into S3 then it returns MD5 of that object.
- One can easily check it with previously computed hash value to guarantee data integrity. Short term availability depends upon the Amazon's connectivity and load on its server at that

instant.

- Once the data is actually in the S3 then it is Amazon's responsibility to take care of it's availability. They claim that the data is backup on multiple hard drives in multiple data centers but doesn't guarantee this by any service Level Agreement.
- There is no backup or recovery mechanism if the user accidentally deletes any data. Amazon has a very impressive scheme of authentication in comparison to other cloud services. Every AWS account has an Access key ID and a Secret key.
- The ID is of 20 characters and the key is a 41 character string. When signing HMAC is first computed for the sign request parameters using that key. And in the Amazon server that HMAC is again computed a compared with the value previously computed in client side. These requests also include timestamp prevent reply attacks.

Elastic Compute Cloud:

- As the name implies EC2 are cloud of computers to the users with flexibility of choosing the configuration of the virtual machine like RAM size, local disk size, propcessor speeds etc.
- Machines that deliver EC2 services are actually virtual machines running on top of XEN platform. Users can machine store a disk image inside S3 and create a virtual machine in EC2 using tools provided by Amazon This virtual machine can be easily instantiated using a java program and can also be monitored. As EC2.

is based on XEN it supports by any linux distribution as well as other OSs. Amazon does not promise about reliability of the EC2 Computers.

- Any machine can crash at any moment and they are not backed up. Although these machine generally don't crash according to the experience of the users but it is safe to use S3 to store information which is more reliable and replicated service. EC2 security model is similar to that of S3. The only difference is that the commands are signed with an X 509 private key. But this key is downloaded from AWS account so the security depends fundamentally on the AWS username and password.

Title:

Case study on microsoft Azure to learn about microsoft Azure is a cloud computing platform and infrastructure, created by microsoft, for building, deploying and managing applications and services through a global network of microsoft managed datacenters. How it work, different services provided by it.

Case study: Microsoft azure.

Abstract:

- Microsoft Azure is a cloud computing service created by Microsoft for building, testing, deploying and managing applications and services through a global network of Microsoft managed data centers. It provides Software as a Service (SaaS) platform as a Service (PaaS) and information as a Service (IaaS) and support many different programming languages, tools and frameworks, including both Microsoft specific and third-party software and systems.
- Azure was announced in October 2008, started with codename "Project Red Dog", and released in February 1, 2010 as "Windows Azure" before being renamed "Microsoft Azure" on March 25, 2014.

KEYWORDS:

- Introduction
- Architecture
- Services

Introduction:

- Microsoft Azure uses a specialized operating system, called Microsoft Azure, to run its its "febric layer": a cluster hosted at microsoft's data centers that manages computing and storage resources of computers and provisions the resources (or a subset of them) to applications running on top of microsoft Azure.
- Microsoft Azure has been described as a "cloud layer" on top of a number of Windows Server systems, Which use Windows Server 2008 and a customized version of Hyper-V known as the microsoft Azure Hypervisor to provide virtualization of services.
- Scaling and reliability are controlled by the microsoft Azure Fabic controller, Which ensures the services and environment do not fail if one or more of the servers fails within the microsoft data center, and which also provides the management of the user's Web application Such as memory allocation and load balancing.
- Azure provides an API built on REST, HTTP, and XML that allows a developer to interact with the services provided by microsoft Azure, microsoft also provides a client-side managed class library that encapsulates the function of interacting with the services with microsoft Visual Studio, Git and Eclipse.

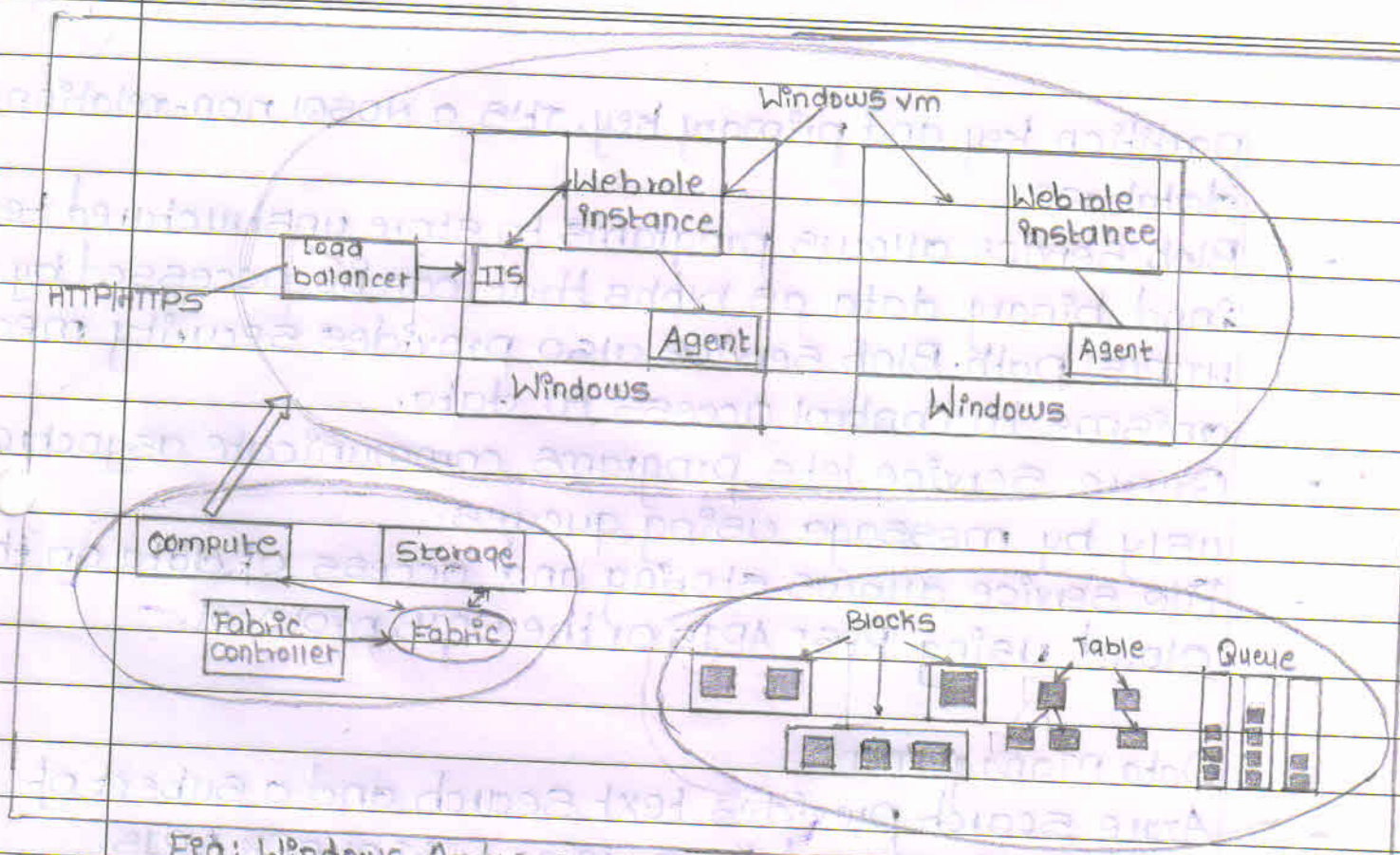

Fig: Windows Azure component Architecture.

Services:

Mobile Services:

- Mobile engagement collects real-time analytics that highlight users' behavior. It also provides push notifications to mobile devices.
- HockeyApp can be used to develop, distribute and beta-test mobile apps.

Storage Services:

- Storage services provides REST and SDK APIs for storing and accessing data on the cloud.
- Table service lets programs store structured text in partitioned collections of entities that are accessed by

partition key and primary key. It's a NOSQL non-relational database.

- Blob Service allows programs to store unstructured text and binary data as blobs that can be accessed by a HTTP(S) path. Blob service also provides security mechanisms to control access to data.
- Queue Service lets programs communicate asynchronously by message using queues.
- File service allows storing and access of data on the cloud using REST APIs or the SMB protocol.

Data Management:

- Azure search provides text search and a subset of OData's structured files using REST or SDK APIs.
- Cosmos DB is a NOSQL database service that implements a subset of file the SQL SELECT statement on JSON documents.
- Redis cache is a managed implementation of Redis.
- StorSimple manages storage task between on-premises devices and cloud storage.
- SQL database, formerly known as SQL Azure database, works to create, scale and extended applications into the cloud using Microsoft SQL server technology. It is also Integrates with Active Directory and Microsoft System Center and Hadoop.
- SQL Data Warehouse is a data warehousing service designed to handle computational and data intensive queries on datasets exceeding 1TB.
- Azure data Factory, is a data integration service that

allows creation of data-driven Workflows in the cloud for orchestrating and automating data movement and data transformation.

- Azure Data lake is scalable data storage and analytic service for big-data analytics Workloads that require developers to run massively parallel queries.
- Azure HDInsight is a big data relevant service that deploys Hortonworks Hadoop on microsoft Azure, and supports the creation of Hadoop clusters using linux with Ubuntu.
- Azure stream Analytics is a serverless scalable event processing engine that enables users to develop and run real-time analytics on multiple streams of data from sources such as devices, sensors, Web sites, social media, and other applications.

Messaging:

- The microsoft Azure Service Bus allows applications running on Azure premises or off premises devices to communicate with Azure. This helps to build people scalable and reliable applications in a service-oriented architecture (SOA). The Azure service bus support four different types of communication mechanisms:
- Even Hubs, Which provide event and telemetry ingress to the cloud at massive scale, with low latency and high reliability. For example an event hub can be used to track data from cell phones such as a GPS location coordinate in real time.
- Queues which allows one-dimentional communication. A sender application would send the message to the servi-

ce but queue, and a receiver would read from the service queue. Though there can be multiple readers for the queue only one would process a single message.

- Topics, which provides one-directional communication using a subscriber pattern. It is similar to a queue; however each subscriber will receive a copy of the message sent to a Topic. Optionally the subscriber can filter out messages based specific criteria defined by the subscriber.
- Relays, which provide bi-directional communication. Unlike queues and topics, a relay doesn't store in-flight messages in its own memory. Instead it just passes them on to the destination application.